BUS 445 Final Exam Cheatsheet: Customer Analytics

1. Core Concepts & Strategy

- **Interpretation First:** Focus on *why* a result matters. Keep answers concise (2-4 sentences).
- **Signal vs. Noise:**
  - **Signal:** Repeatable patterns in data.
  - **Noise:** Unique, non-repeatable random fluctuations.
  - **Overfitting:** When a model captures noise instead of just the signal, leading to poor predictions on new data.
- **Measurement Scales:**
  - **Nominal:** Categories with no order (e.g., Region, Gender). Use Mode.
  - **Ordinal:** Ordered categories (e.g., Low/Med/High, Satisfaction). Use Median/Mode.
  - **Interval:** Numeric, no "true zero" (e.g., Temperature in C). Use Mean/SD.
  - **Ratio:** Numeric with "true zero" (e.g., $ Spend, Age). Use Mean/SD.

2. Data Preparation & Exploration

- **Binning:** Converting a continuous variable into an ordinal one by creating ranges. Useful for contingency tables or visualizing non-linear relations.
- **Missing Values:**
  - `variable.summary()` shows % NA.
  - Trees/Random Forests handle missing values automatically.
  - Linear/Logistic regression require imputation or removal.
- **Contingency Tables:** Used for two categorical variables.
  - **Expected Value (Null Hypothesis):** $E = \frac{(\text{Row Total} \times \text{Column Total})}{\text{Grand Total}}$
  - **Chi-square p-value:** If $p < 0.05$, reject the null hypothesis (there *is* a relationship).

3. Major Predictive Methods

Linear Regression (`lm`)

- **Interpretation:** "For every 1 unit increase in X, Y changes by [Estimate], holding others constant."
- **Dummy Variables:** A nominal variable with N levels is represented by $N-1$ dummy (0/1) variables. The missing level is the **reference/baseline**.
- **Non-linearity:** If a U-shape exists, include both X and X^2 in the model.
- **Multicollinearity:** When predictors are highly correlated, individual p-values may be misleading.

Logistic Regression (`glm`)

- **Target:** Binary (0/1, Yes/No).
- **Logit Formula:** $\ln(\frac{p}{1-p}) = \beta_0 + \beta_1 X_1 + \dots$
- **Odds:** $\frac{p}{1-p} = e^{(\beta_0 + \beta_1 X_1 + \dots)}$
- **Probability:** $p = \frac{e^{\text{logit}}}{1+e^{\text{logit}}}$
- **Interpretation:** Coefficients represent change in *Log-Odds*. Positive = higher probability.

Decision Trees (`rpart`) & Random Forest (`randomForest`)

- **Pros:** Easy to explain (Trees), handle non-linearity and missing values automatically.
- **Cons:** Trees can be unstable; Random Forests are "black boxes" but highly accurate.
- **Pruning:** Reducing tree size to avoid overfitting.

Neural Networks (`Nnet`)

- **Pros:** Can capture extremely complex, non-linear relationships.
- **Cons:** Hardest to interpret; requires tuning (Size, Decay).
- **Size:** Number of hidden nodes. **Decay:** Penalty to prevent weights from getting too large (prevents overfitting).

4. Model Assessment & Targeting

Lift Charts

- **Baseline:** The horizontal line representing random targeting (expected response rate).

- **Incremental Response:** Response rate in each specific group (e.g., top 10%).
- **Cumulative Response:** Total percentage of all responders captured by targeting the top $X\%$.
- **Targeting Strategy:** Target customers in deciles where the expected profit > cost of contact.

Overfitting & Validation

- **Estimation Sample:** Used to build the model.
- **Validation Sample:** Used to check for overfitting and compare models.
- **Oversampling:** Used when the target event is rare (e.g., 1% response). It helps the model see enough "Yes" cases.
  - **Calibration:** Probabilities from an oversampled model must be adjusted back to the "true" population rate using `adjProbScore()`.

5. Grouping Methods

Cluster Analysis (Vertical Summary)

- **Goal:** Group similar *individuals* (rows) into segments.
- **K-Means:** Minimizes distance to cluster **mean**.
- **K-Medians:** Minimizes distance to cluster **median** (robust to outliers).
- **Validation:**
  - **Adj. Rand Index:** Measures stability (closer to 1 is better).
  - **C-H Index:** Measures "tightness" vs. "separation" (higher is better).

Principal Components Analysis (Horizontal Summary)

- **Goal:** Group similar *variables* (columns) into "components".
- **Usage:** Used to reduce dimensions for plotting clusters or handling multicollinearity.

6. Key R Functions Reference

Task	Function	Package/Source
Summary	`variable.summary(df)`, `numSummary(df)`	`BCA_functions.R`
Binning	`binVariable(x, bins=4, method="proportions")`	`BCA_functions.R`
Samples	`create.samples(df, est=0.5, val=0.5)`	`BCA_functions.R`
LinReg	`model <- lm(target ~ x1 + x2, data=train)`	`stats`
LogReg	`model <- glm(target ~ x1 + x2, family=binomial, data=train)`	`stats`
NNet	`Nnet(target ~ x1 + x2, data=train, size=3, decay=0.1)`	`BCA_functions.R`
Assessment	`lift.chart(modelList, data=val, targLevel="Yes", trueResp=0.05)`	`BCA_functions.R`
Scoring	`rawProbScore()`, `adjProbScore(..., trueResp=0.05)`	`BCA_functions.R`
Effects	`plot(allEffects(model))`	`effects`
Clustering	`bootCVD(data, k=2:6, method="kmn")`	`BCA_functions.R`

7. Simple Calculations for Exam

1. **Probability from Odds:** $P = \frac{\text{Odds}}{1+\text{Odds}}$
2. **Odds from Probability:** $\text{Odds} = \frac{P}{1-P}$
3. **Lift Calculation:** $\text{Lift} = \frac{\text{Model Response Rate}}{\text{Baseline Response Rate}}$
4. **Expected Value (Contingency):** $\frac{\text{Row Total}\times\text{Col Total}}{\text{Grand Total}}$